PCA:

You are given 4 data samples, each of dimension 3. The data matrix reads:

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}$$

The eigenvectors of $\mathbf{X}^T\mathbf{X}$ are:

$$\mathbf{u}_1 = \tfrac{1}{\sqrt{3}}(1,1,1)^T$$
$$\mathbf{u}_2 = \tfrac{1}{\sqrt{2}}(-1,0,1)^T$$
$$\mathbf{u}_3 = \tfrac{1}{\sqrt{2}}(-1,1,0)^T$$

These eigenvectors are already normalized and they are already correctly ordered corresponding to eigenvalues in descending order. (Just ignore that the second and third eigenvector are not orthogonal. This is due to the fact that the data matrix only has rank 1.)

Compute the PCA projection (matrix) $\mathbf{Y}$.

Enter the components of the projected second feature vector $\mathbf{y}_2$ into the fields below.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $y_2^{(1)} =$
- $y_2^{(2)} =$
- $y_2^{(3)} =$

Hierarchical Clustering

You are given the set of data points: $\mathbf{x} = \{a, c, e, h, i, m, n, s\}$,

where the similarity between two points $s(x, x')$ is defined as the distance between the ordered letters in the alphabet,
i.e. $s(a, b) = 1$, $s(a, c) = 2$ and so on.

Perform agglomerative clustering of the data using average linkage -
the distance between clusters $A$ and $B$ is defined as $d_{avg}(A, B) = \frac{1}{n_A} \frac{1}{n_B} \sum_{a \in A} \sum_{b \in B} s(a, b)$.

Stop when the distance between clusters is greater than 7.

Answer the following questions (enter integer values):

- How many clusters are formed in total?
- What is the maximum number of data points in any one cluster?
- What is the minimum distance between any two of the formed clusters?

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct?

Multiple answers are possible!

Select one or more:

- [ ] a. In the M-step, the hidden variables are updated.
- [ ] b. Maximum Entropy requires minimal prior assumptions to assign a-priori-probabilities.
- [ ] c. The E-step increases the upper bound on the log-likelihood.
- [ ] d. In the E-step, the model parameters are updated.
- [ ] e. The EM algorithm can be applied to hidden Markov models and mixture models.
- [ ] f. The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- [ ] g. The E-step increases the lower bound on the log-likelihood.
- [ ] h. In EM, one deals with problems that have hidden variables as well as model parameters.

Assume you have two random variables $Y$ and $Z$, which are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively, i.e.

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y} & \text{if } x \in [0, \theta_Y] \\ 0 & \text{else.} \end{cases}$$

$$p_Z(x) = \begin{cases} \frac{1}{\theta_Z} & \text{if } x \in [0, \theta_Z] \\ 0 & \text{else.} \end{cases}$$

For $\theta_Y = e^5$ and $\theta_Z = e^{10}$, compute the KL divergence

$$D_{\mathrm{KL}}(p_Y || p_Z) = \int_{-\infty}^{\infty} p_Y(x) \, \log[p_Y(x)/p_Z(x)] \, \mathrm{d}x.$$

As for "log", use the natural logarithm.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $D_{\mathrm{KL}}(p_Y || p_Z) =$ ☐.

Which statements about Scaling and Projection Methods are correct?

Multiple answers are possible!

Select one or more:

- ☐ a. Multidimensional scaling (MDS) keeps the distances between data points (as well as possible).
- ☐ b. The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- ☐ c. Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space.
- ☐ d. In t-SNE, the similarity between data points $i$ and $j$ is given by their joint probability.
- ☐ e. In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ f. The objective in t-SNE is minimization of the KL divergence between the similarities of the (high-dimensional) inputs and the (low-dimensional) mapped outputs.
- ☐ g. In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.
- ☐ h. LLE in general performs linear mappings to realize neighborhood-preserving embeddings.

Assume you want to do $k$-means on the following 2-dimensional dataset:

$$X = \{(1,-1),(0,0),(-1,-1),(0,-2),(2,0),(2,3),(5,0)\}.$$

The initial cluster centers are $\boldsymbol{\mu} = (0,0)$ and $\mathbf{v} = (2,1)$.

Compute the coordinates of the new cluster centers $\boldsymbol{\mu}^{\text{new}} = (\mu_1^{\text{new}}, \mu_2^{\text{new}})$ and $\mathbf{v}^{\text{new}} = (\nu_1^{\text{new}}, \nu_2^{\text{new}})$ after one interation of $k$-means.

Hint: Make a sketch first. The problem can mainly be solved graphically.

- $\mu_1^{\text{new}} =$ ☐, $\mu_2^{\text{new}} =$ ☐
- $\nu_1^{\text{new}} =$ ☐, $\nu_2^{\text{new}} =$ ☐

Mixture model:

Assume the following case:

The prior probability of component $j$ is 0.1.

The probability for observing a data sample $\mathbf{x}$ given the total parameter set $\boldsymbol{\theta}$ is 0.2.

The likelihood for observing a data sample $\mathbf{x}$, given component $j$ and corresponding parameter vector, is 0.5.

Compute the posterior probability of component $j$.

You can use a pocket calculator. State your final result with two decimal places. Number format example: 0.67

- Posterior = ☐

Which statements about Independent Component Analysis (ICA) are correct?

Multiple answers are possible!

Select one or more:

- ☐ a. For ICA, one needs more sources than observations.
- ☐ b. Maximizing the negentropy maximizes the mutual information.
- ☐ c. Criteria for independence are mutual information between components and non-Gaussianity of components.
- ☐ d. The ICA solution is unique.
- ☐ e. Fast ICA relies on whitening and rotating the data.
- ☐ f. Infomax maximizes the mutual information between components.
- ☐ g. Non-Gaussianity can be measured by the second cumulant.
- ☐ h. In general, ICA components are neither ranked nor orthogonal.

Assume you are given two binary random variables $X$ and $Y$ taking only the values $0$ and $1$ where their joint probability distribution is specified by the table below:

| | $Y = 0$ | $Y = 1$ |
|---|---|---|
| $X = 0$ | 1/3 | 1/3 |
| $X = 1$ | 0 | 1/3 |

This means, that e.g. $P(X = 0, Y = 0) = 1/3$, etc.

Compute the quantities (marginal probabilities, entropies, and mutual information) below. Use the natural logarithm in your calculations.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $P(Y = 0) =$ ☐
- $H(X) =$ ☐
- $H(Y) =$ ☐
- $I(X; Y) =$ ☐

For any parameter $w$, an estimator $\hat{w}$ is ____ if $\mathbb{E}(\hat{w}) = w$. The ____ of $\hat{w}$ can be decomposed as a sum of its ____ and its squared ____ .

Let $X_1, X_2, \ldots, X_n$ be random variables drawn independently from a distribution $p(\cdot; w)$. Consider $\frac{1}{n+1} \sum_{i=1}^{n} X_i$ as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is ____ .

Now consider $X_1$ as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is ____ .

Let $\{x_1, x_2, \ldots, x_n\}$ be a sample drawn independently from the distribution $p(\cdot; w)$. The function $\prod_{i=1}^{n} p(x_i; w)$ is called the ____ function. Since $p(\cdot; w)$ is usually a ____ function, the probability measure of the set $\{x_1, x_2, \ldots, x_n\}$ is in fact zero.

The Cramer-Rao lower bound relates the ____ of the estimator $\hat{w}$ to the inverse of its Fisher information. An estimator that reaches the Cramer-Rao lower bound is said to be ____ .